# BSc (Hons) in Information Technology

## Year 3

## Assignment 1

**IT3040 – ITPM** **Semester 1**

### Option 2: Functional and Usability Testing of a Website (for students not familiar with Sinhala)

This assessment should be attempted **individually**. The assessment is worth **18%** of the overall marks for the unit. This assignment is designed to evaluate your ability to test a live system, design automation strategies, and analyze its weaknesses in a structured manner.

### ➢ Objective:

- The main objective of this assignment is to assess the **functional correctness** and **usability** of the website available at https://www.pixelssuite.com/ by verifying that its features behave as expected and evaluating whether the **user interface** is **clear**, **consistent**, and **easy to use** under normal operating conditions. Note that **backend API testing**, **performance testing**, **scalability testing**, and **security testing** are **outside the scope of this assignment**.

- Accordingly, you are required to test following **ten** features of the website:
  1) Document conversion
  2) PDF editing
  3) Image resizing
  4) Cropping
  5) Compression
  6) Image format conversion
  7) Meme generation (available under 'More' option)
  8) Color picker (available under 'More' option)
  9) Image rotation (available under 'More' option)
  10) Image flipping (available under 'More' option)

### ➢ Tasks:

- Identify **thirty-six** test scenarios, ensuring that **at least three** scenarios are designed to test each of the **ten** features listed above. For each feature, the **three** scenarios must include **two negative** test cases, in which the feature is evaluated under **invalid or unexpected** conditions, and **one positive** test case, in which the feature operates correctly as intended under **valid** conditions. The **remaining six** scenarios may be designed to test any of the features listed above.

- From the identified scenarios, **automate one** test scenario to verify the **preview functionality** of **any one** of the features listed **below** using an image file in **PNG** format. Use **Playwright** to record the execution results of this automated scenario in the provided CSV file titled "**execution_results**." The steps that should be followed for automating a scenario using Playwright are provided in the "**Automation Steps - Option 2**" document. In addition, the "**test_automation_ui**" folder contains the files required to run Playwright and generate the results.
  - Image resizing
  - Cropping (Crop PNG)
  - Compression (PNG Compressor)
  - Image format conversion

- The remaining **thirty-five scenarios** should be recorded **manually** using the **test case template** provided in the **Excel file** titled "***Manual Test Cases for Option 2***." Refer to **Appendix 1** for guidance on how to complete the Excel file.

## ➢ Required Files:

1. The **full Playwright project repository** (including all scripts and configuration files), along with the **Git repository link** provided in a separate **text file**. The repository should include clear instructions in a **README.md** on how to install dependencies and run the tests. Ensure that the Git repository is **publicly accessible**. Repositories that cannot be accessed during marking will **not** be evaluated.
2. The completed **CSV file**, titled "**execution_results.**"
3. The completed **Excel file**, titled "**Manual Test Cases for Option 2**."

**Plagiarism Notice:**

Submitted files will be checked for plagiarism. Any submission with a similarity score **greater than 10%** will be considered plagiarized, and **no marks will be awarded** for the assessment.

## ➢ Method of Submission:

- Rename all files with your registration number.
- Create a folder with your registration number
- Paste all the required files into the folder.
- Zip the folder.
- Upload the zipped folder to the *'**Assignment 1 Answer: Option 2**'* link available on CourseWeb **before 5th May**.

# Appendix 1

### Sample Test Case Template

| TC ID | Application Feature Tested | Input | Expected output | Actual output | Status | Assumption for Expected Output |
|---|---|---|---|---|---|---|
| Neg_0001 | Image resizing | Upload a non-image file like .txt | The system should reject the file upload and display a message indicating that the file type is not supported. | The system rejected the file upload, but no message was displayed to inform the user of the issue. | Fail | It is assumed that the system should display a validation message when an unsupported file type is uploaded. |
| Pos_0002 | Image resizing | Upload a valid PNG image | The uploaded image should be shown in the Preview section | The uploaded image was displayed correctly in the Preview section | Pass | NA |

**Note:**

1. The Test Case (TC) IDs of **positive** test cases should begin with "**Pos**", while the IDs of **negative** test cases should begin with "**Neg.**"
2. If any assumptions are made regarding the **expected output**, they should be documented in the "***Assumption for Expected Output***" column.